

Karthikheyaa Kurra

267-299-9245 | karth@udel.edu | [GitHub](#) | [Portfolio](#) | [LinkedIn](#)

EDUCATION

University of Delaware

M.S. Data Science, GPA 3.80/4.00

Newark, DE

Aug 2024 – May 2026

- **Coursework:** Probability Theory, Mathematical Techniques for Data Science (Optimization), Applied Multivariate Statistics, Machine Learning, Algorithm Design & Analysis, Research Software Engineering

Sathyabama Institute of Science & Technology

B.E. Computer Science & Engineering

Chennai, India

Oct 2020 – May 2024

EXPERIENCE

Research Associate, Scientific Machine Learning

Jun 2026 – Present

Department of Mathematical Sciences, University of Delaware (Advisor: Dr. Ke Chen)

Newark, DE

- Writing a paper on hybrid PDE solvers that combine classical methods and neural networks to solve nonlinear elliptic PDEs non-iteratively via learned surrogates and hierarchical structure.
- Built high-precision ground truth with Chebyshev spectral discretizations and Newton iteration, plus dataset validation and reproducible PyTorch training/evaluation pipelines.
- Ran ablation studies on network depth, normalization, and physics-informed residual losses with adaptive loss balancing; profiled inference complexity and wall-clock time.
- Maintained the research codebase with unit/smoke tests and weekly progress reports.

Research Software Engineer, LLM Serving & Benchmarking

Jun 2026 – Present

Department of Computer & Information Sciences, UD (Advisor: Dr. Sunita Chandrasekaran)

Newark, DE

- Part of a team building an on-premise LLM assistant for UD students, faculty, and staff.
- Benchmarking LLM inference/serving (throughput, latency, model selection) on NVIDIA DGX hardware.

Research Assistant, Scientific Computing

Jun 2025 – Jun 2026

Department of Mathematical Sciences, University of Delaware (Advisor: Dr. Ke Chen)

Newark, DE

- Built a hierarchical Poincaré–Steklov (HPS) solver for linear elliptic PDEs: Chebyshev leaf grids, Dirichlet-to-Neumann maps, and Schur-complement merges up the tree.
- Verified on manufactured solutions (Laplace, Poisson, Helmholtz) and wrote notes on leaf operators, merges, and hierarchy scaling.

Teaching Assistant, Machine Learning

Jan 2024 – Apr 2024

Sathyabama Institute of Science & Technology

Chennai, India

- Graded assignments, midterms, and homework; prepared lecture slides and notes for the undergraduate ML course.
- Led review sessions on mathematics for machine learning (matrix algebra and multivariable calculus).

Research Intern, Computer Vision

Mar 2023 – Jun 2023

Indian Institute of Technology, Hyderabad

Hyderabad, India

- Applied supervised contrastive learning (Khosla et al., 2020) to facial expression recognition in PyTorch, comparing SupCon against a cross-entropy baseline.
- Analyzed embeddings with t-SNE and delivered a reproducible codebase to the lab.

Summer Intern

Jun 2022 – Jul 2022

The University of Texas at Dallas

Dallas, TX

- Summer program in machine learning and deep reinforcement learning: lectures, labs, and model building from first principles.

TECHNICAL SKILLS

Languages: Python, MATLAB, R, SQL, Bash, LaTeX

Scientific ML: PyTorch, NumPy/SciPy, Operator Learning, Physics-Informed Neural Networks (PINNs)

Tools: Git, Docker, Linux, HPC job scripting, VS Code, Cursor

Reading Group: SciML Reading Group, University of Delaware (Advisor: Dr. Ke Chen), 2025–Present